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Collar type differential which combines limited slip and locking capability

Abstract

A differential assembly includes a first case half. A second case half is connected to the first case half. The first and second case halves define a gear chamber. The first and second case halves each include pockets formed therein. A differential gear set is disposed in the gear chamber. First and second clutch packs are disposed in the pockets between the first and second case halves and differential gear set. The clutch packs engage the first and second case halves and the differential gear set defining a limited slip. A locking collar is disposed in a cavity formed in one of the first or second case halves. An electric actuator mechanism is coupled to the locking collar selectively moving the locking collar into engagement with the differential gear set locking the differential.

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Background/Summary

FIELD OF THE INVENTION

(1) The invention relates to differentials and more particularly to limited slip locking differentials.

BACKGROUND OF THE INVENTION

(2) A conventional limited slip differential (LSD) cannot give full locking capability. Further, prior art locking differentials are usually noisy and the tires may wear out quickly. There is therefore a need in the art for better handling and off-road capability associated with a LSD and allow complete locking when desired.

SUMMARY OF THE INVENTION

(3) In one aspect, there is disclosed a differential assembly includes a first case half A second case half is connected to the first case half. The first and second case halves define a gear chamber. The first and second case halves each include pockets formed therein. A differential gear set is disposed in the gear chamber. First and second clutch packs are disposed in the pockets between the first and second case halves and differential gear set. The clutch packs engage the first and second case halves and the differential gear set defining a limited slip. A locking collar is disposed in a cavity formed in one of the first or second case halves. An electric actuator mechanism is coupled to the locking collar selectively moving the locking collar into engagement with the differential gear set locking the differential.

(4) In another aspect, a differential assembly includes a first case half A second case half is connected to the first case half. The first and second case halves define a gear chamber. The first and second case halves each include pockets formed therein. A differential gear set is disposed in the gear chamber. First and second clutch packs are disposed in the pockets between the first and second case halves and differential gear set. The clutch packs engage the first and second case halves and the differential gear set defining a limited slip. A locking collar is disposed in a cavity formed in one of the first or second case halves. An electric actuator mechanism is coupled to the locking collar selectively moving the locking collar into engagement with the differential gear set locking the differential. One of the first or second case halves includes a laterally extending wall formed thereon. The laterally extending wall separates the clutch pack from the locking collar.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a section view of a differential gear mechanism;

(2) FIG. 2 is an enlarged view taken from FIG. 1.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(3) Referring to FIG. 1, the differential gear mechanism **10** includes a first or right gear case **12** and a second or left gear case **14**, which defines gear chamber **16**. Torque input to the differential is typically provided by a ring gear (not shown), which may be attached to a flange **18** of the right gear case **12** or the left gear case **14**.

(4) Disposed within the gear chamber **16** is a differential gear set including a pair of input pinion gears **18**, which are rotatably mounted on a pinion shaft **20**. The pinion gears **18** are in meshing engagement with a pair of side gears **22** and **24**.

(5) As is well known to those skilled in the art, during normal straight-ahead operation of the vehicle, no differentiation occurs between the left and right side gears **22** and **24**, and therefore, the pinion gears **18** do not rotate relative to the pinion shaft **20**. The gear case halves **12**, **14**, the pinion gears **18**, and the side gears **22** and **24** all act as a unit. When the vehicle is turning differentiating action may occur between the side gears **22** and **24**.

(6) A pair of clutch packs or first and second clutch packs **26** are positioned in pockets **28** between the case halves **12**, **14**, and the opposing side gears **22**, **24**. The clutch **26** packs include a plurality of clutch disks **30**, which are in splined engagement with the gear cases **12** and **14**. The clutch **26** packs also include a plurality of clutch disks **32**, which are in splined engagement with the side gears **22**, **24**. The clutch packs **26** provide a limited slip differential capability or define a limited slip.

(7) The clutch packs include a preload spring which applies some pressure on the clutch pack **26** and more pressure is applied when the engine power/torque is applied to the side gears **22**, **24** through the pinion gears **18**. When torque is applied to pinion gears **18**, this creates an axial force, which tries to force the pinion gears **18** apart. This in turn puts pressure on the side gears **22**, **24** forcing it outward toward the differential case halves **12**, **14**. This outward force compresses the clutch disks **30**, **32** together causing them to lock.

(8) When the vehicle is moving straight ahead, the differential acts like a standard differential, as the clutch packs **26** are not engaged. When turning or when one wheel has low traction, the wheel with the most traction has higher turning resistance than the wheel, which is slipping. The engine torque applied through the pinion gear **18** to the side gear **22**, **24** with higher traction causes more separating force between the pinion gear **18** and side gear **22**, **24**. This forces the side gear **22**, **24** toward the differential case halves **12**, **14** engaging the clutch pack **26**. The side gear **22**, **24** of the slipping wheel has little turning resistance to the torque. It does not have the same force on the side gear toward the differential case halves **12**, **14** and thus does not compress the clutch disks **30**, **32**.

(9) The differential gear mechanism **10** includes a direct acting electronic locking capability. An electric actuator mechanism **34** includes at least one actuating rod **36** coupled to an electric stator **38**. Various numbers of stators **38** and actuating rods may be utilized. The at least one actuating rod **36** in the depicted embodiment passes through the right case half **12** in a bore and is coupled to a locking collar **40**. The locking collar **40** includes teeth **42** formed on an inner diameter of the locking collar, **40** which selectively engage with the outer diameter of the side gear **24** to lock the differential. The locking collar **40** is disposed within a cavity **44** defined by the gear case halves **12**, **14** and the side gear **24**. The locking collar **40** is biased to a disengaged position by a biasing spring **46**. In the depicted embodiment the biasing spring **46** biases the locking collar toward the right gear case.

(10) In the depicted embodiment, the left case half **14** includes a laterally extending wall **48** formed thereon. The laterally extending wall **48** separates the clutch pack **26** from the locking collar **40**. The laterally extending wall **48** includes a notch **50** formed thereon that receives one end of the spring **46** with the opposing end contacting the locking collar **40**.

(11) In use, the differential gear mechanism **10** includes clutch packs **26** on both side gears **22**, **24** to provide a limited slip functionality as described above. The electric actuator mechanism **34** is normally biased to a disengaged position by the spring **46**, which moves the locking collar **40** away from engagement with the side gear **24**. Upon activation of the stator **38**, the locking collar **40** is moved by the actuating rod **36** to engage with the side gear **24** to lock the differential gear mechanism **10**. The actuation may be in response to a control signal from a control module or may be actuated in response to activation of a switch by a user. After the stator is de-energized, the differential gear mechanism **10** is returned to a disengaged position by the spring **46**. The above described differential gear mechanism **10** provides a collar type direct acting differential which combines limited slip and locking capability.

Claims

1. A differential assembly comprising: a first case half; a second case half connected to the first case half, the first and second case halves defining a gear chamber, the first and second case halves each including pockets formed therein; a differential gear set disposed in the gear chamber; a first

clutch pack and a second clutch packs disposed in the pockets between the first and second case halves and differential gear set, the first and second clutch packs engaging the first and second case halves and the differential gear set defining a limited slip; a locking collar disposed in a cavity formed in one of the first or second case halves; and, an electric actuator mechanism coupled to the locking collar selectively moving the locking collar into engagement with the differential gear set locking the differential; wherein the locking collar includes teeth formed on an inner diameter of the locking collar.

2. The differential assembly of claim 1, wherein the electric actuator mechanism includes at least one actuating rod coupled to an electric stator on one end and coupled to the locking collar on an opposing end.

3. The differential assembly of claim 2, wherein the one of the first or second gear case halves includes a bore formed therein, the at least one actuating rod disposed therein, wherein the electric stator is positioned on one of the gear case halves and the locking collar is positioned in the other gear case half.

4. The differential assembly of claim 1, wherein one of the first or second case halves includes a laterally extending wall formed thereon, the laterally extending wall separating the first clutch pack from the locking collar.

5. The differential assembly of claim 4, wherein the laterally extending wall includes a notch formed therein.

6. The differential assembly of claim 5, further including a biasing spring having one end disposed in the notch and an opposing end contacting the locking collar biasing the locking collar to an unlocked position.

7. The differential assembly of claim 1, wherein the differential gear set includes a pair of input pinion gears rotatably mounted on a pinion shaft, the pinion gears in meshing engagement with a pair of side gears.

8. The differential assembly of claim 1, wherein the teeth engage an outer diameter of a side gear when in a locked position.

9. The differential assembly of claim 1, wherein the first and second clutch packs each include a plurality of clutch disks in splined engagement with the first and second case halves.

10. The differential assembly of claim 9, wherein the first and second clutch packs each include a plurality of clutch disks in splined engagement with side gears.

11. A differential assembly comprising: a first case half; a second case half connected to the first case half, the first and second case halves defining a gear chamber, the first and second case halves each including pockets formed therein; a differential gear set disposed in the gear chamber; a first clutch pack and a second clutch packs disposed in the pockets between the first and second case halves and differential gear set, the first and second clutch packs engaging the first and second case halves and the differential gear set defining a limited slip; a locking collar disposed in a cavity formed in one of the first or second case halves; and, an electric actuator mechanism coupled to the locking collar selectively moving the locking collar into engagement with the differential gear set locking the differential, wherein one of the first or second case halves includes a laterally extending wall formed thereon, the laterally extending wall separating the first clutch pack from the locking collar, wherein the laterally extending wall includes a notch formed therein and including a biasing spring having one end disposed in the notch and an opposing end contacting the locking collar biasing the locking collar to an unlocked position.

12. The differential assembly of claim 11, wherein the first and second clutch packs each include a plurality of clutch disks in splined engagement with the first and second case halves and wherein the first and second clutch packs each include a plurality of clutch disks in splined engagement with side gears.

13. The differential assembly of claim 11, wherein the locking collar includes teeth formed on an inner diameter of the locking collar wherein the teeth engage an outer diameter of a side gear when

in a locked position.

14. A differential assembly comprising: a first case half; a second case half connected to the first case half, the first and second case halves defining a gear chamber, the first and second case halves each including pockets formed therein; a differential gear set disposed in the gear chamber, the differential gear assembly comprising a side gear with an outer diameter; a first clutch pack and a second clutch pack disposed in the pockets between the first and second case halves and the differential gear set, the first and second clutch packs engaging the first and second case halves and the differential gear set defining a limited slip; a locking collar disposed in a cavity formed in one of the first or second case halves; an electric actuator mechanism coupled to the locking collar selectively moving the locking collar into engagement with the differential gear set locking the differential; and wherein the locking collar selectively engages with the outer diameter of the side gear to lock the differential.
